

Evaluation of the cooling effectiveness of air-cooled data centers by energy diagram

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HIGHLIGHTS

- Four phases of first cooling system of data centers are presented, and the energy analysis diagram is developed.
- The new index of effective cooling rate (ECR) is proposed to evaluate the efficiency of the first cooling system.
- Three modes are simulated, and the results show that the enclosed aisle is more efficiency with a high value of ECR.
- ECR can provide a new method to couple a three-dimensional CFD simulation with a one-dimensional energy consumption model.

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ABSTRACT

Data centers are one of the world's largest energy consumers, and their cooling systems contribute to the majority of the total energy consumption. Although some studies have suggested evaluation metrics, the cooling capacity utilization in the computer room of air-cooled data centers is still not able to be directly quantified. In this paper, the dimensionless index Effective Cooling Ratio (ECR) is proposed for the first time to evaluate the energy efficiency of the cooling process in air-cooled data centers. By decoupling the cooling process of the computer room, the energy diagram is developed to analyze the thermal energy changes of each phase. Moreover, the proving of this approach is carried out with computational fluid dynamics simulation; three data center models are investigated, and the results demonstrate the good performance of the enclosed aisle. This metric not only provides a valuable tool for quantifying the cooling and energy performance within data centers but also suggests ways to improve the energy efficiency of data centers.

1. Introduction

Data centers were born and developed along with the booming information technology industry, and they provide physical support for fields such as big data, cloud computing, and artificial intelligence. Driven by the accelerating growth of computing capability, the energy consumption of data centers is ever-increasing. Masanet et al. [1] estimated that the global data center energy consumption in 2018 was 205 TWh, and the International Energy Agency suggested that in 2022, it was 240–340 TWh [2], which accounts for about 1–1.3 % of global energy consumption, and it predicts a double power consumption of data centers by 2026 [3]. Energy consumption in data centers is mainly for IT equipment and infrastructure support systems. The former is the data center's productivity, and the latter provides the operating

conditions required by the IT equipment, such as cooling, lighting, power supply, and so on. Among them, the cooling system is acknowledged to be the most energy-consuming part [4].

The target of data center cooling is to cool down the electronic devices. Take the air-cooled data center as an example; the heat is generated by the working IT devices and carried away by the airflow. The hot air transfers heat to a cooling medium inside the coil of the heat exchanger, and finally, the heated cooling medium transfers the heat to the environment [5]. The above-mentioned cooling process is composed of two cooling systems: the first cooling system and the second cooling system, as shown in Fig. 1. In the first cooling system, mainly convective heat transfer occurs between hot electronic devices and air, which is unique to data centers, while in the second cooling system, heat transfers from the hot medium to the environment, and it is common to large

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energy-consuming buildings. The two cooling systems are interactive, and their efficiencies determine the energy consumption of the cooling system [6].

Therefore, a better understanding of energy use in the first cooling system is important. To reveal this issue, information on air velocity and temperature distributions in the data center is needed. For any data center, access is strictly restricted, and any activity that may pose a risk is not permitted. These limitations virtually rule out the possibility for researchers to conduct detailed measurements of air velocity and temperature in the computer room. Thus, Computational Fluid Dynamics (CFD) modeling has become the main approach to carrying out such studies and has been widely adopted in the past decades. The following is a brief review of this.

Schmidt et al. [7] simulated a small section of an IBM raised-floor data center and studied the various open ratios of perforated tiles to make almost uniform flow rates. Karki et al. [8] used commercial CFD software TileFlow to discover the effect of plenum height and underfloor partitions. Shrivastava et al. [9] modeled various arrangements of air supply and return on the ceiling, the diffuser, the sidewall, and the tiles. Nada and Said [10] investigated the temperature distribution of the underfloor airflow parallel and perpendicular to the cold aisle by changing the location of the computer room air conditioning (CRAC) and found the perpendicular plan showed better results. Gondipalli et al. [11] employed the professional simulation software FloTHERM to compare the cases with several kinds of cold aisle to the base case without enclosure and found that “solid doors and a slit on roof with offset” is the best choice to optimize the air-stream organization of a data center. Shrivastava et al. [12] studied a data center of high power density, with the room sizes being 23.2 m × 29.9 m and a total of 135 racks of two powers: 6.7 kW and 19 kW. Their results were verified by experiments.

According to the simulated results, how to evaluate the effectiveness of cooling is a research priority, which can guide the direction for improvement. Some representative indicators are summarized in Table 1. These evaluation indicators can be broadly divided into two kinds: temperature & airflow evaluation and energy evaluation.

(1) Temperature & airflow evaluation: Sharma et al. [13] proposed two indices called the Supply Heat Index (SHI) and The Return Heat Index (RHI) in 2002 to evaluate hot and cold airflow mixing. The former indicates the enthalpy rise of the hot infiltration divided by the total enthalpy rise at the rack outlet, and the latter shows the ratio of the total extraction of CRACs to the total enthalpy rise at the rack outlet. Obviously, the sum of SHI and RHI equals 1. Schmidt et al. [14] suggested index β in 2005 to measure the local inefficiency. It uses the temperature difference to evaluate the self-heating loop. Herrlin [15] introduced rack cooling index (RCI) to compare the cooling effect with industry guide or product standard. It provides recommended and allowable temperatures

Table 1
Some indicators and definitions.

No.	Index	Definitions
1	SHI and RHI	$SHI = \frac{\delta Q}{Q + \delta Q} = \frac{\sum_i \sum_j ((T_{in}^R)_{ij} - T_{out}^C)}{\sum_i \sum_j ((T_{out}^R)_{ij} - T_{out}^C)}$ $RHI = \frac{Q}{Q + \delta Q} = \frac{\sum_k m_k^C ((T_{in}^C)_k - T_{out}^C)}{\sum_i \sum_j m_{ij}^R ((T_{out}^R)_{ij} - T_{out}^C)}$ $SHI + RHI = 1$
2	β	$\beta = \frac{T_{in}^R - T_{out}^C}{T_{out}^R - T_{in}^R}$
3	RTI	$RTI = \frac{T_{return} - T_{supply}}{\Delta T_{equipment}} \times 100\%$
4	RCI	$RCI_H = \left[1 - \frac{\sum_n ((T_x)_n - T_{rec})}{(T_{all} - T_{rec})n} \right] \times 100\%$ $RCI_L = \left[1 - \frac{\sum_n (T_{rec} - (T_x)_n)}{(T_{rec} - T_{all})n} \right] \times 100\%$
6	CI	$CI = \frac{m_{spe}}{m_{tot}}$
7	ELI	$ELI = \frac{\Delta E_{loss}}{\Delta E_{sys}}$
8	EPM	$EPM = \frac{\Psi_{VR}}{\Psi_{AE}}$
9	AMI	$AMI = \frac{\Delta R_{sys} - \Delta R_{nm}}{\Delta R_{sys}}$
10	PUE and pPUE _{cooling}	$PUE = \frac{P_{total}}{P_{IT}} = \frac{P_{cooling}}{P_{IT}}$ $PUE \approx 1 + pPUE_{cooling}$
11	COPUE	$COPUE = \frac{PUE_{real}}{PUE_{bm}}$
12	PUC	$PUC = \frac{P_{total}}{C_{total}}$

for upper and lower limits, and can help to identify whether the cooling supply is excessive or insufficient. Return temperature index (RTI) is used to capture the utilization of the temperature rise, which assesses the performance of airflow management by whether the bypass and recirculation exist. Vangilder and Shrivastava [16] presented the capture index (CI) in 2007, which only focuses on the performance of airflow pattern. These metrics quantify the temperature distribution and airflow organization, and hence evaluate the cooling impact within a data center.

(2) Energy evaluation: Fakhim et al. [17] in 2012 developed an exergetic performance metric (EPM) to evaluate the exergy destruction of the cold aisle, hot aisle, ceiling region and the region between rack and CRAC wall. Qian et al. [18] in 2013 used air mixing index (AMI) based on entransy dissipation method to analyze the thermal resistance of hot and cold air mixing. Xie et al. [19] raised exergy loss index (ELI) in 2019 to assess the proportion of the exergy loss of the total exergy, and the loss is irreversible which includes recirculation flow, bypass flow

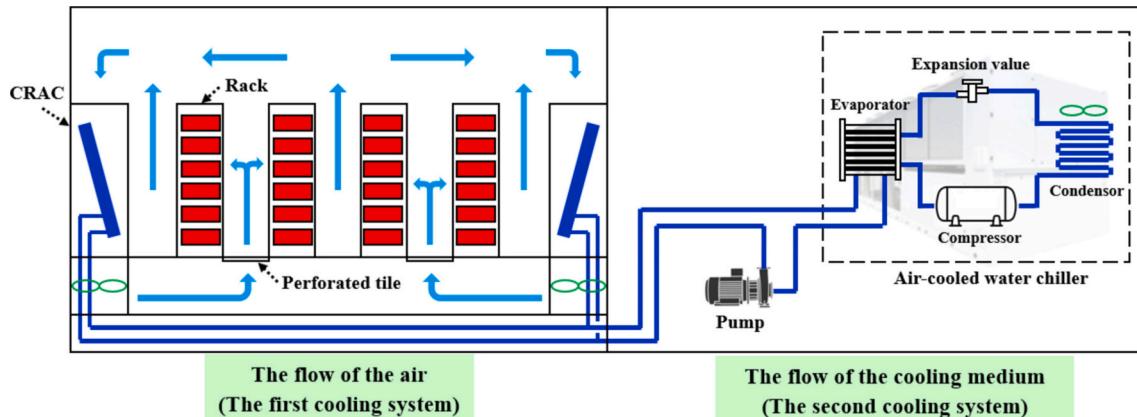


Fig. 1. Two cooling systems.

and negative pressure flow. These evaluation methods have employed different theoretical approaches and investigate the degree energy utilization of air cooling in data centers.

The green grid proposed the Power Usage Effectiveness (PUE) [20] to evaluate the energy efficiency of data center operations and is widely used in many countries and organizations. It is defined by the ratio of the total energy consumed over the power of IT devices. The total energy consumption of the data center is composed of the energy consumption of the IT equipment, cooling system, electrical system losses, and other energy consumption. Apart from PUE, partial PUE (pPUE) is used to evaluate the energy utilization of these individual parts. Since the cooling system accounts for the majority of the energy consumption of infrastructure, the value of the PUE of the data center can be approximately equal to one plus the partial PUE of the cooling system (pPUE-cooling). In 2020 Li et al. [21] proposed a new indicator for a fair comparison on the energy performance of data centers which can compensate for differences due to climate impacts. Li et al. [22] in 2024 further explored the energy efficiency based on data center output, that is, the power consumption of computing, which enables feasible evaluation metrics including IT performance, and established a unified benchmark evaluation methodology for computing data centers. These evaluation methods examine the overall data center performance and are constantly modified and adapted in accordance with different data center structure.

However, even though more than 20 indices for evaluating the data center air-cooling performance have been proposed [23], no one directly related to the energy efficiency of the first cooling system, and it is still difficult to evaluate through these indices. With the authors' knowledge, the existing indices or metrics cannot meet such requirements, and in some papers, even though modern mathematics theories are adopted, the obtained results are still far from that requirement. Advantages your technique brings and why this is important. In this paper, we innovatively propose an energy diagram representation of the air cooling process in data centers, through which the temperature, flow rate, and energy changes can be obtained for each phase of the data center cooling, and the interaction relationship among them. The energy diagram integrates the temperature (y-axis), airflow (x-axis) and energy (area) evaluation, and provides a visual and intuitive analysis method.

Then based on the energy diagram, the dimensionless index Effective Cooling Ratio (ECR) is proposed, which realizes to evaluate the energy efficiency of the first cooling system, that is, the amount of cooling capacity applied to the IT equipment rack. The index of ECR fill in the missing link of the energy efficiency evaluation system for data center, from the evaluation for IT equipment cooling, cooling system electrical efficiency and overall energy efficiency of the data center. In addition, this index can provide a combination of one-dimensional system-level energy simulation and three-dimensional room-level environment simulation, thereby helping to improve the full-scale data center digital twin model in future research.

The rest of the paper is organized as follows. In Section 2, the numerical model for this study is introduced, and in Section 3, the grid independence analysis and validation are presented. Then, in Section 4, the four phases of the first cooling are discussed in detail, and the energy diagram and effective cooling rate (ECR) are proposed. In Section 5, the numerical results of three airflow organization structures are described. Finally, some conclusions are provided in Section 6.

2. Physical and numerical models of an air-cooled data center

The most important requirement for any index is its feasibility and applicability to a practical data center. Based on this understanding, this study adopted a physical model of the air-cooled data center cell from Rambo and Joshi [24], which has the layout of a common design pattern for underfloor-air-supply data centers. The space of the data center is $11.68\text{m} \times 9.43\text{m} \times 3\text{m}(\text{h})$, and the plenum is 0.6m deep. The racks ($0.61\text{m} \times 0.76\text{m} \times 2\text{m}(\text{h})$) lie in 4 rows, with seven racks in each row

and an interval of 2 perforated tiles ($1.22\text{m} \times 1.22\text{m}$, 25 % open area). The top view and the front view of the data center cell are shown in Fig. 2. The rack consists of 10 servers, the heat load per server is 423W , and the fan at the server provides cold air approximately $480\text{m}^3/\text{h}$ (from the curve of Papst series 6400 axial fan). The CRAC ($3\text{m} \times 0.9\text{m} \times 1.8\text{m}(\text{h})$) is installed at the middle of each wall, and the supply air is 15°C at the rate of $5.7\text{m}^3/\text{s}$. The cold air is blown into the underfloor plenum from the bottom of the CRAC and sent up to the room through the perforated tiles; then, it is drawn towards the racks to cool down the IT equipment. Finally, the heated air flows back to the return inlet at the top of the CRAC.

The main heat balance in a first cooling system is dominated by the cooling load of air handling units and the heat generated from racks, while the heat gain/loss from the external enclosure and the heat generated by the other equipment are of a small proportion and can be ignored in the study. The simulation is conducted with the following assumptions: (1) The airflow in the data center is incompressible and turbulent. (2) The process is in steady state. (3) The rack servers' heat load is the only heat source. (4) Radiation heat transfer is neglected. (5) Air thermal physical properties are constant. (6) The building enclosure is assumed to be adiabatic and has no leakage from or to the environment.

The numerical model is governed by the mass, momentum, and energy conservation equations, as shown in Eqs. (1) ~ (3).

Mass conservation equation:

$$\nabla \cdot (\rho \mathbf{u}) = 0 \quad (1)$$

Momentum conservation equations:

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot ((\mu_t + \mu_r) [\nabla \mathbf{u} + \nabla \mathbf{u}^T]) + S \quad (2)$$

S is source term, including gravity and porous resistance force.

Energy conservation equation:

$$\nabla \cdot (\rho c_p \mathbf{u} T) = \nabla \cdot ((\lambda_t + \lambda_r) \nabla T) \quad (3)$$

where the turbulent transport parameter μ_t , λ_t are determined by the $k-\varepsilon$ turbulent model [25,26].

The finite volume method is used for discretization. The convective terms are discretized by second-order upwind scheme, diffusion terms

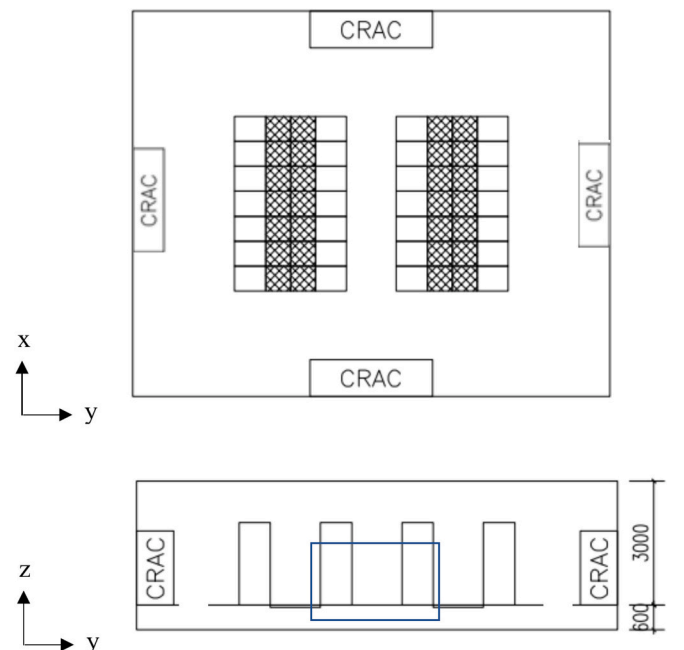


Fig. 2. The top view and the front view of the data center cell.

by second-order central scheme, and the coupling between velocity and pressure is solved by the SIMPLE algorithm. The commercial CFD software ANSYS Icepak is employed in this study.

The numerical rack model is an important element in the data center simulation. Two kinds of simplified models are widely used in the room-level study: the black-box model [26] and the porous-media model [27] (Fig. 3). In the black-box model, the rack region is treated as a non-computational domain, and its air inlet and outlet are regarded as domain outlet and inlet with the same airflow rate. The temperature of the rack's back side is calculated based on the average temperature of the front side plus the temperature difference obtained by the ratio of the server's heat load divided by the heat capacity of the flow rate. The porous medium model of the rack considers the space inside the rack as the porous medium and is simulated in conjunction with the space outside the rack. As a result, the whole space of the first cooling region is the computational domain, and the heat generation of the real server is transferred to the air flowing over the server. The airflow rate and the pressure drop through the rack are determined by the equipment embedded in the rack. The porous medium model is more realistic; hence, it is applied in this study.

For the simulation of the porous medium region, the built-in porous medium model in the Icepak is adopted, where the required porous medium parameters l_{c1} , l_{c2} are determined from the data in [24]. In this study, we have fitted several sets of data in [24] on pressure drop and flow rate to identify the loss coefficients of porous media model (Eq. 4).

$$\Delta p = C_1 + C_2 = \frac{\rho}{2} l_{c1} v + \frac{\rho}{2} l_{c2} v^2 \quad (4)$$

3. Grid independence study and modeling validation

Three grid systems are generated in Icepak, and the locations of monitoring points are shown in Fig. 4. The results of the simulated temperature of the three grid systems are shown in Table 2, also the GCI by Roache criteria [28] are calculated. It can be seen that the GCI of the middle size grid is already very close to 0. Considering the balance between accuracy and required computer sources, the grid number of around 2 million is adopted in the following study.

The temperature field clouds at $z = 1.5$ m of this study are shown in Fig. 5, and the comparison of the temperatures at the monitoring points between the present study and reference [24] is shown in Table 3.

The local deviations for points 6, 7, 9, and 12 in Table 3 are relatively large, and they may be caused by following possible reasons: 1) modeling errors; 2) technological inefficiency when taking data from literature figures. The points 6 and 12, and points 7 and 9 are respectively symmetrical. Generally speaking, the temperature deviations are within an acceptable range for practical study [29]. The validation shows that this modeling approach of CFD simulation is feasible for data center research, and it will be used in the following works.

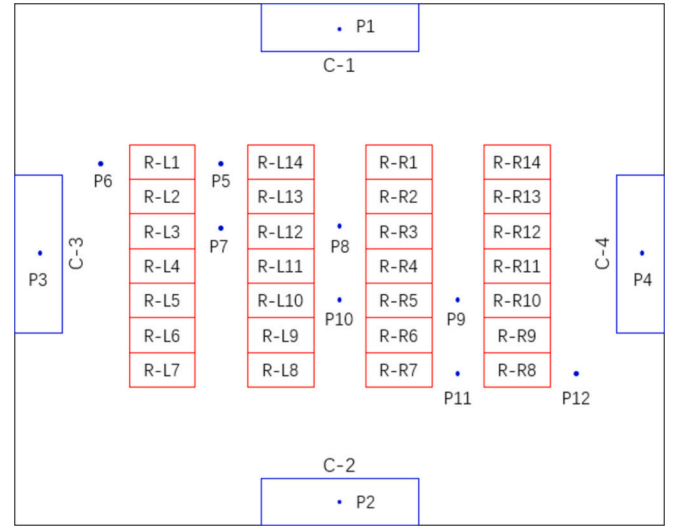


Fig. 4. The locations of the monitoring points.

Table 2

The temperature of monitoring points.

Monitoring points	Grid number		
	336,016	1,852,299	4,131,232
1	18.95°C	18.94°C	19.00°C
2	18.96°C	18.95°C	18.99°C
3	19.81°C	19.96°C	20.05°C
4	19.77°C	19.98°C	20.05°C
5	18.42°C	18.35°C	18.81°C
6	20.41°C	20.44°C	20.73°C
7	16.61°C	16.56°C	16.64°C
8	17.03°C	17.07°C	17.04°C
9	16.55°C	16.56°C	16.65°C
10	17.03°C	17.07°C	17.03°C
11	18.41°C	18.36°C	18.80°C
12	20.49°C	20.46°C	20.79°C
Roache criteria	–	0.023 %	0.065 %*

Note: * Compared to the second set of grid.

4. Heat transfer and effective cooling rate

This section explores the heat transfer process in air-cooled data centers and develops an energy diagram to quantify the heat transfer visually. Finally, an effective cooling rate is proposed to evaluate the heat transfer in the first cooling system.

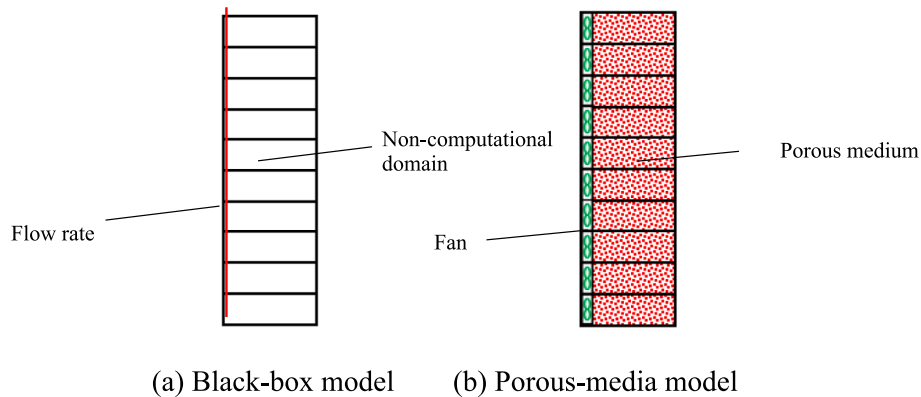


Fig. 3. Schematic of two rack models.

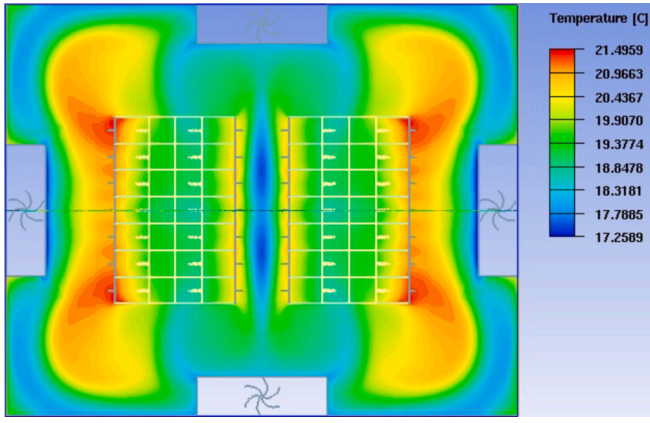


Fig. 5. The temperature field at $z = 1.5$ m of this study.

Table 3
The comparison of the temperatures at the monitoring points.

Monitoring point	Temperature in this study /°C	Temperature in Rambo and Joshi's work /°C	Relative deviation
5	18.42	17.7	4.07 %
6	20.41	18.4	10.92 %
7	16.61	17.8	-6.69 %
8	17.03	17.1	-0.41 %
9	16.55	17.8	-7.02 %
10	17.03	17.1	-0.41 %
11	18.41	17.7	4.01 %
12	20.49	18.5	10.76 %

4.1. Four phases of the first cooling system

The cooling process of air-cooled data center can be divided into four phases based on the heat transfer as shown in Fig. 6. Four phases of the first cooling system consist of the cooling air delivery, the cooling action, the exhaust of hot air, and the heat removal by CRAC coils.

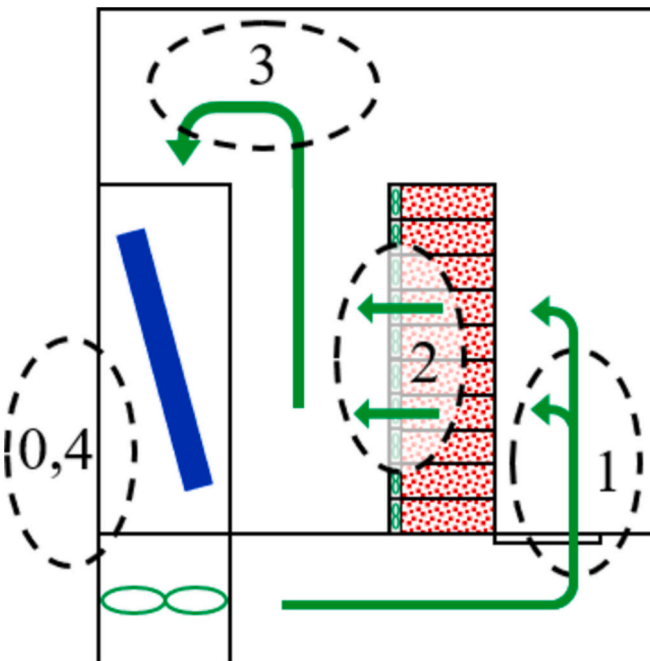


Fig. 6. Four phases of the ideal cooling process.

- (1) The first phase is the cooling air delivery, i.e., the cold air from CRAC is delivered into the rack inlet, such as underfloor air supplied by CRAC flows via perforated tiles. The so-called recirculation may occur in this phase, which means part of the heated air returns back to the rack inlet, mixes with the cold air, and thus increases the rack inlet temperature. According to the analysis made in [30], the internal server temperature, such as chip temperature, is positively correlated with the rack inlet temperature. Therefore, the hot air infiltration will intensify the local hot spots.
- (2) The second phase is the cooling action; that is, the air stream is induced by server fans through racks to cool down the electronic components. At a steady state, this processing will always carry away the same amount of heat as the power consumption of IT equipment.
- (3) The third phase is the exhaust of hot air, i.e., the hot air out of the rack is organized through the room-level ventilation and returned to the CRAC. In this phase, there may be a part of the cold air that is not engaged in the previous phase and is mixed up in this local region, which is called bypass. It raises the rack outlet temperature higher than the design temperature. Because of this bypass, the flowrate of the cold air that goes through the rack is reduced, thereby leading to the increase in rack outlet temperature for a fixed rack power according to the energy balance equation:

$$Q = mC(t_{out} - t_{in}) \quad (5)$$

where m —air flow rate, m^3/s ; $C = \rho c_p$, $J/(m^3 \cdot K)$. c_p is the specific heat capacity of air, $J/(kg \cdot K)$; ρ is air density, kg/m^3 .

(0) The last phase, and also the beginning of another cycle, is the removal of heat given by rack servers by CRAC coils, and then the new cooling air starts another cycle. It is important that this phase makes the air-cooling process sustainable, which serves as the interface of the first and second cooling systems.

As discussed above, there may be some unwanted air flows in the four phases, and for multiple racks, the schematic of four phases with some unwanted flows is schematically shown in Fig. 7. The existing evaluation metrics may relate to different phases. For example, the degree of perfection of Phase-1 is commonly evaluated using the SHI index. Obviously, its ideal value is 0: the larger the value, the worse the utilization of the cooling capacity, and more recirculation occurs, and non-zero SHI means the actual cooling air temperature of the rack is higher than the CRAC's set point. On the other hand, the RTI index of 100 % indicates that the cooling output of the air condition is appropriately assigned to each server at Phase-2, and a value of RTI bigger than 100 % means a larger temperature difference between the supply and return air than the designed one, hence more cooling air is required.

Even though SHI and RTI may describe the perfectness of Phases 1

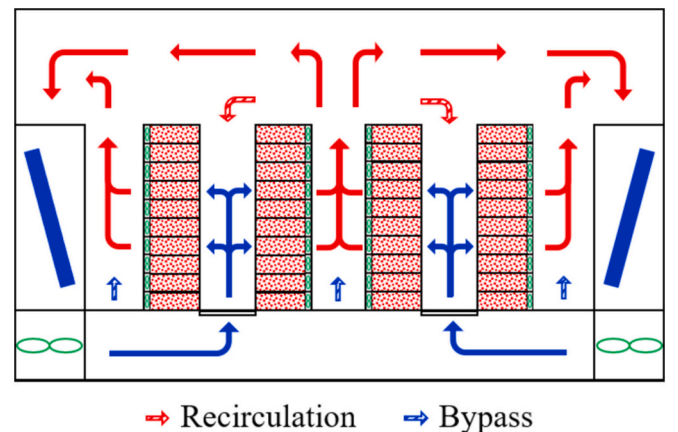


Fig. 7. Schematic of four phases in air-cooled data center.

and 2, generally speaking, it is still difficult to quantify the percentage of the cooling capacity of the cold air directly applied to the IT equipment, which is going to be discussed below.

4.2. The energy analysis diagram

For the benefit of quantitative discussion, Fig. 7 is represented by a flowchart, as shown in Fig. 8, where the recirculation and bypass between the hot and cold aisles are clearly expressed. In addition, the interface is the contact surface of two successive phases. They reflect the states of the beginning and end of two successive phases, and the heat exchange between the interfaces indicates the heat exchanges in the phase.

This paper proposes a diagram method to analyze the heat exchange with a two-dimensional coordinate system, where the value of horizontal coordinate x is proportional to the flow rate m , and the vertical coordinate y is the weighted average temperature T of the interface. In the figure, every point at the x -coordinate has the following flow rate:

$$x = 2 C m \quad (6)$$

To show one phase in the diagram, we mark two points of interface-a and interface-b in the same vertical line as shown in Fig. 9 and then make the projection of point interface-a onto the y -axis, named interface-a'. Here, the vertical coordinate of interface-a represents the average temperature of the phase inlet, while that of interface-b is the outlet temperature. A triangle is thus formed by "interface-a, interface-a', interface-b"; its area is calculated by Eq. (6) and equals the amount of heat exchange within this process. In this way, the proposed diagram can be used to visually analyze the energy exchanges for each phase. For the convenience of discussion, this diagram is called the energy diagram later.

$$A = \frac{1}{2} \cdot x \cdot \Delta T = c_p m \cdot (T_b - T_a) \quad (7)$$

where A —area of triangle, W .

ΔT —temperature difference between a and b, $^{\circ}C$.

T_a —temperature of a, $^{\circ}C$.

T_b —temperature of b, $^{\circ}C$.

By putting the four phases together in the energy diagram, in the order of "Phase-1, Phase-2, Phase-3, Phase-0", a complete heat transfer process for the first cooling system of the data center of the ideal condition can be obtained as shown in Fig. 10. Obviously, the triangle areas of Phase-2 and Phase-0 are the same, and the heat exchanges are taken in rack and CRAC. It should be noted that the x -coordinate of each stage starts from zero.

The expressions of recirculation, bypass, and their combination in the energy diagram are respectively demonstrated in Figs. 11 (a), (b), (c). For the recirculation, with the fixed IT load, a new thermal energy

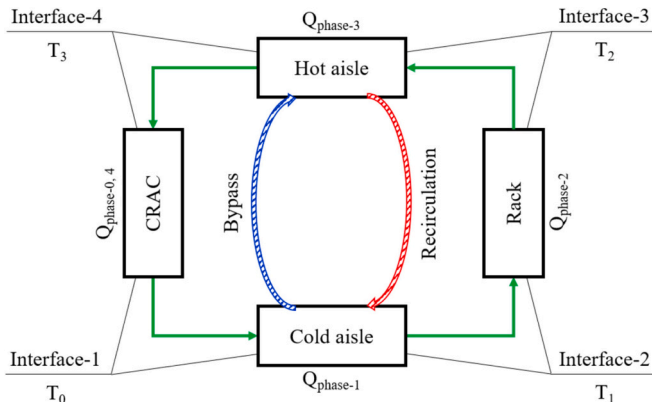


Fig. 8. Flowchart for four phases.

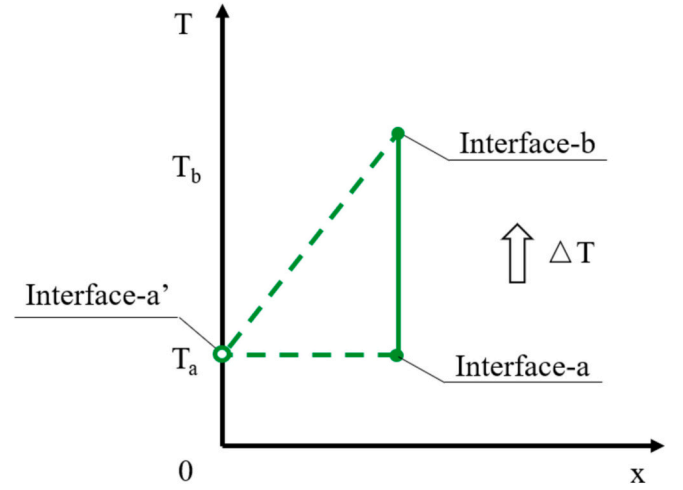


Fig. 9. Explanation of energy diagram

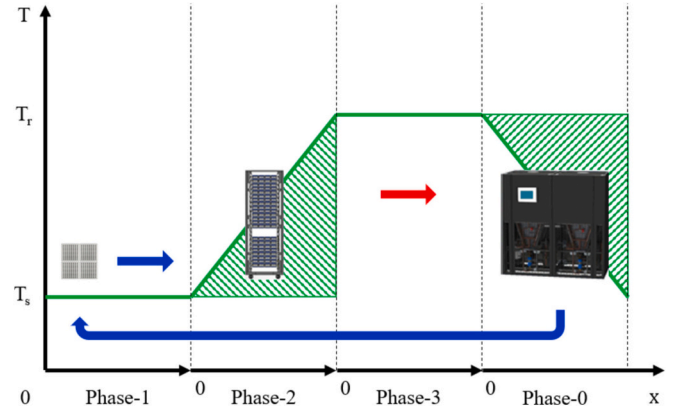


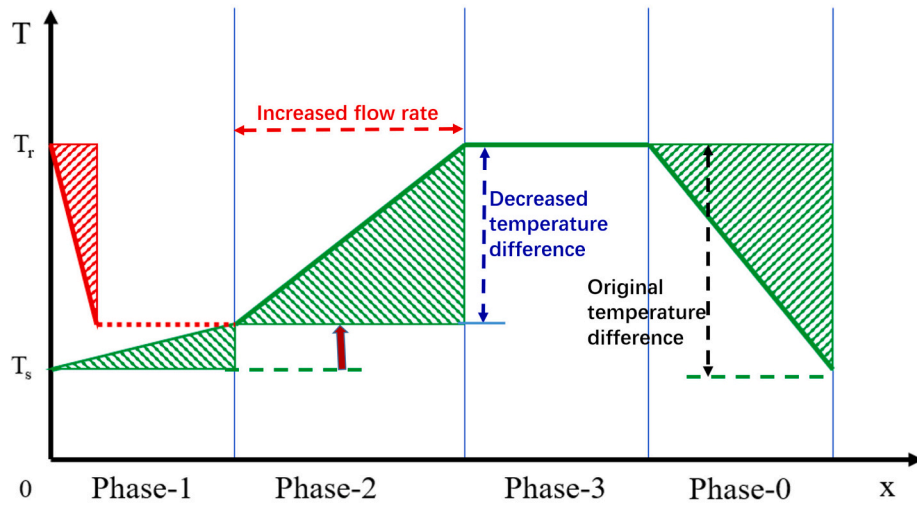
Fig. 10. Multiple energy diagram of four phases under ideal conditions.

balance of the cooling air deviating from the design conditions should be re-established. Since the temperature at the rack inlet is increased, the temperature difference between the IT surface and the cooling air stream becomes less; hence, the flow rate within Phase-2 becomes larger. The decrease in temperature difference and increase in flow rate have been clearly shown in Fig. 11 (a). For the bypass, the flow rate decreases within Phase-2, and the temperature at the rack outlet is raised. However, the air mixes in the room and restores to the design temperature at the air conditioning return. This creates a temperature trick that can be seen in Fig. 11 (b). For the combination, it's a blend of recirculation and bypass, with all the negative effects. As a side note, because these two phenomena occur simultaneously, the inlet and out temperature are both increased, while the airflow in Phase-2 may be close to the design condition, as illustrated in Fig. 11 (c).

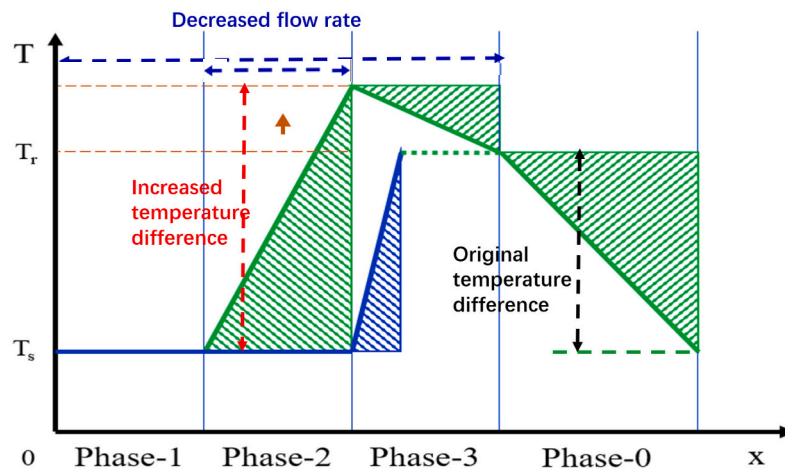
From the diagram, it can be obtained that all temperatures are lifted, although the temperature of the air conditioning remains unchanged, more cooling capacity is needed in order to regain the design conditions, and the specific adjustments include lowering the supply air temperature or/and adding the flow rate of cooling air. All these will increase the energy consumption of the cooling system of the data center. Therefore, the ECR is very important, and it can directly affect the energy efficiency of the data center.

4.3. The effective cooling ratio

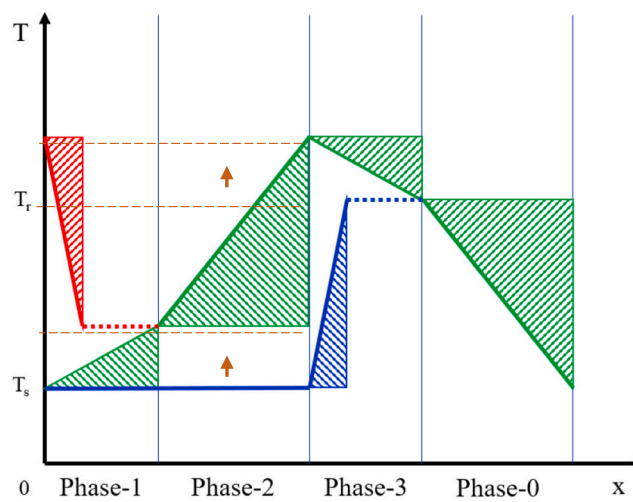
The effective cooling ratio (ECR) is the percentage of cooling capacity that is directly attributable to the IT equipment. From the above



(a) Energy diagram with recirculation



(b) Energy diagram with bypass



(c) Energy diagram with combination

Fig. 11. Energy diagrams with recirculation and bypass.

diagrams, we can see that the heat exchange (the area of the triangle) is consistent in Phase-2 regardless of conditions, whereas it does change in Phase-1 and Phase-3. Therefore, through assessing the ratio of heat exchange wasted in Phase 1 and Phase 3 to total heat exchange within Phase 2, the inverse is related to the effective cooling ratio (ECR) as follows:

$$ECR = 1 - \frac{Q_{\text{phase-1}} + Q_{\text{phase-3}}}{Q_{\text{phase-2}}} \quad (8)$$

where $Q_{\text{phase-n}}$ —the heat exchange within phase n ($n = 1, 2, 3, 0$), W.

In combination with the energy analysis diagram, the ECR calculation can be converted into the intuitive area ratio, as shown in Eq. (9).

$$ECR = 1 - \frac{A_{\text{phase-1}} + A_{\text{phase-3}}}{A_{\text{phase-2}}} \quad (9)$$

where $A_{\text{phase-n}}$ —the area of triangle of phase n , W.

In the energy diagram the vertical coordinate is the temperature, and the horizontal coordinate about the flow rate m . The inflow-outflow relationships between major mass flow and the bypass/recirculation of air stream for rack and CRAC are shown schematically in Fig. 12. The quantitative relationship of airflow organization for recirculation and bypass is shown in Eqs. (10) and (11).

$$m^C = m_{\text{out-net}}^C + m_{\text{byp}} = m_{\text{out-net}}^R + m_{\text{byp}} \quad (10)$$

where m^C —the total air flow rate of CRAC, m^3/s ;

$m_{\text{out-net}}^C$ —the net flow rate of cold air from CRAC direct to rack, m^3/s ;

m_{byp} —the flow rate of bypass airflow, m^3/s ;

$m_{\text{out-net}}^R$ —the net flow rate of hot air from rack direct to CRAC, m^3/s .

$$m^R = m_{\text{out-net}}^C + m_{\text{rec}} = m_{\text{out-net}}^R + m_{\text{rec}} \quad (11)$$

where m^R —the total air flow rate of rack, m^3/s ;

m_{rec} —the flow rate of recirculation, m^3/s .

The energy diagram for the case with bypass and recirculation is presented in Fig. 13.

For Phase-1, the heat exchange is as follows.

$$A_{\text{phase-1}} = \sum_i \sum_j (A_{\text{rec}})_{ij} \quad (12)$$

$$A_{\text{rec}} = cm_{\text{rec}} \cdot (T_{\text{out}}^R - T_{\text{in}}^R) = cm_{\text{out-net}}^C \cdot (T_{\text{in}}^C - T_{\text{out}}^C) \quad (13)$$

For the Phase-2, the heat exchange is as following.

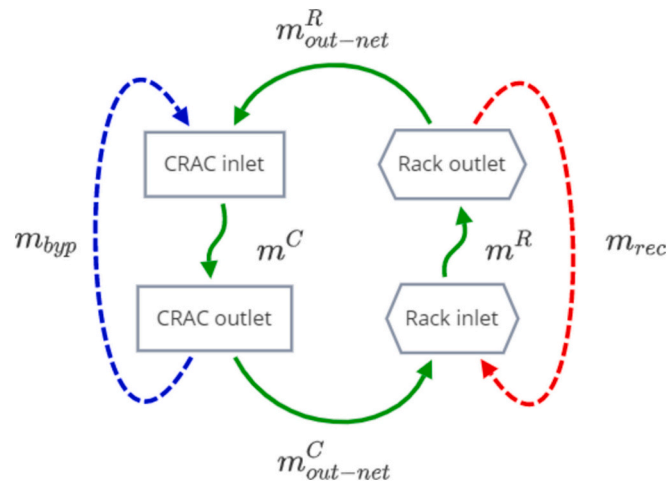


Fig. 12. The airflow organization and flow rate of each part

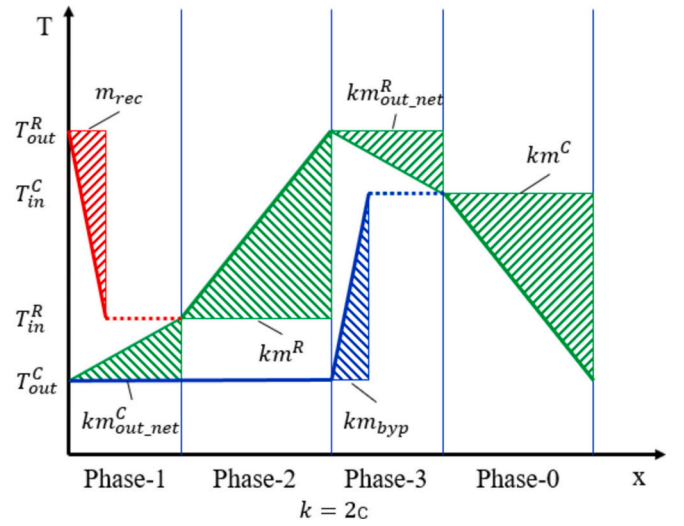


Fig. 13. The energy diagram for the case with bypass and recirculation.

$$A_{\text{phase-2}} = \sum_i \sum_j [cm^R \cdot (T_{\text{out}}^R - T_{\text{in}}^R)]_{ij} \quad (14)$$

For the Phase-3, the heat exchange is as following.

$$A_{\text{phase-3}} = \sum_n (A_{\text{byp}})_n \quad (15)$$

$$A_{\text{byp}} = cm_{\text{byp}} \cdot (T_{\text{in}}^C - T_{\text{out}}^C) = cm_{\text{out-net}}^R \cdot (T_{\text{out}}^R - T_{\text{in}}^C) \quad (16)$$

For the Phase-0

$$A_{\text{phase-0}} = \sum_n [cm^C \cdot (T_{\text{in}}^C - T_{\text{out}}^C)]_n \quad (17)$$

Then

$$ECR = 1 - \left[\frac{\sum m_{\text{rec}} (T_{\text{out}}^R - T_{\text{in}}^R)}{\sum m^C (T_{\text{in}}^C - T_{\text{out}}^C)} + \frac{\sum m_{\text{byp}} (T_{\text{in}}^C - T_{\text{out}}^C)}{\sum m^R (T_{\text{out}}^R - T_{\text{in}}^R)} \right] \quad (18)$$

For the ideal operation, there is neither the recirculation nor bypass at Phase-1 and Phase-3, so the value of ECR is equal to 100 %; that is, all the cooling capacity occurs inside the rack to reduce IT equipment temperatures. As the ECR is smaller than 100 %, unwanted heat exchanges occur at Phases-1 and Phase-3, and the amount of cooling capacity actually delivered to IT equipment is decreased. Considering the practical values of COP and PUE at the present circumstance, the ECR are greater than zero. In summary, the ECR can be referred to as the energy efficiency of the first cooling system.

5. Results and discussion

Comparative studies were conducted to quantify the effects of mixing hot and cold airflows in recirculation and bypass processes, and three organization models of airflow were simulated with the same IT load condition. These three models are open aisle, enclosed cold aisle, and enclosed hot aisle, respectively [31]. The schematic of each model is shown in Fig. 14. Fig. 14 (a) is the open aisle; the hot and cold air in the space is free-flowing and mixing without any barriers. Fig. 14 (b) is the enclosed cold-aisle, where the cold air forms a cooling pool, and it can only access the room after being heated through the rack. Fig. 14 (c) is the enclosed hot aisle, where the hot air is sealed and returned directly to the air conditioning. The basic CFD model is presented in Section 2, with 4230 W per rack, an air supply temperature of 15 °C from the air conditioning, and the heat transfer through the enclosed aisle and into the room is ignored.

Fig. 15 displays the simulated temperature field of the horizontal plane. The simulations were conducted at the same given conditions

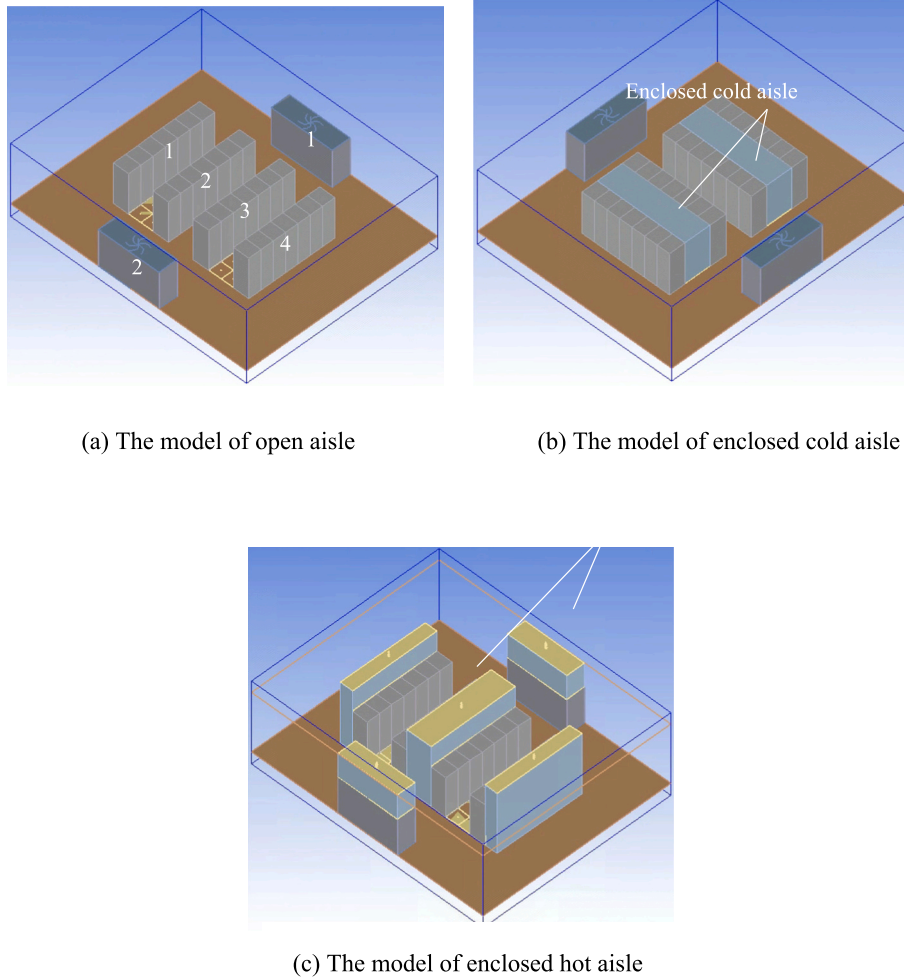


Fig. 14. The schematic of three models.

stated in the 1st paragraph of Section 2. From the figures, it can be seen that the room air temperature of the open aisle model is well mixed, with the average air temperature at the rack inlet being much higher than the temperature of the CRAC outlet, while the temperature distributions of the enclosed aisle are clearly separated. Tables 4–6 provide more detailed information. The return air temperatures of CRAC or CRAC outlet in the three models remained the same, but the rack temperature of the open aisle is different from that of the enclosed ones: its inlet temperature is much higher than the return temperature of CRAC, while the temperature at the outlet did not rise by the same amount. These suggest that the airflow from the rack outlet has partially returned back to the rack inlet, and the airflow through the rack is much increased. The output variables are calculated based on the simulation results by Eqs. (8)–(18).

Note: The serial number of the equipment is marked in Fig. 14(a).

Based on the above numerical data, Table 7 shows four indexes (ECR, SHI, β and RTI) for the three models. The index ECR of open aisle is 37.98 %, and SHI, β , and RTI are equal to 0.6473, 1.8438, and 268 %, respectively. According to the definition of each index, all of them point out the poor performance of the open aisle structure. The index ECR of the enclosed cold aisle and hot aisle is as high as 96.22 % and 89.64 %, respectively, which shows that the physical isolation of the airflow can significantly improve the utilization rate of cooling capacity.

Due to the cold air leakage at the return vent of air conditioning, the effect of hot isolation is slightly worse than cold isolation. In addition, for the hot aisle model, the whole room is considered as a larger cooling pool than the cold aisle formed, and it can provide a comfortable

working condition for maintenance staff and longer operation time in case of cooling failure.

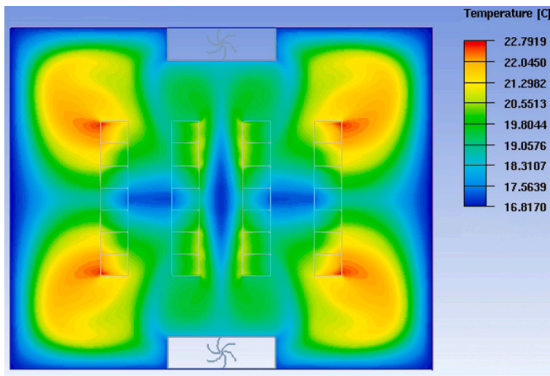
Fig. 16 shows the value of the four indexes. From the figure, it can be seen that the indexes RTI and β are more sensitive to model temperature variations, and the values of SHI are nearly opposite to ECR. However, from their original definitions of the four indices, only the ECR can directly indicate the goodness of cooling capacity utilization.

This study focusses on the energy utilization efficiency of air cooling. Compared with SHI and RHI, they only evaluate the temperature distribution. SHI reflects the mixing of hot and cold air through the temperature variation on the inlet side of the rack, while RHI depicts the mixing on the outlet surface of the rack. The true value of energy efficiency of the first cooling system cannot obtain by SHI or RHI. Even though the values of SHI and RHI can qualitatively reflect whether there is mixing between hot and cold airs, but they cannot be used as a quantitative index.

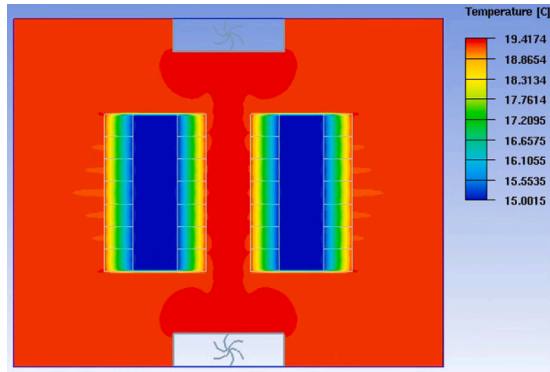
In order to ensure the stable operation of the data center, it is imperative to meet the cooling requirements of IT equipment. The cooling effectiveness can be expressed by PUE and the electric cooling efficiency COP of the second cooling system, and the $pPUE_{cooling}$ of the cooling system is closely related to COP [32] and the effective cooling rate ECR of the first cooling system, as shown in Eq. (19).

$$COP = \frac{\text{the total cooling capacity}}{\text{the input electric power}} \quad (19)$$

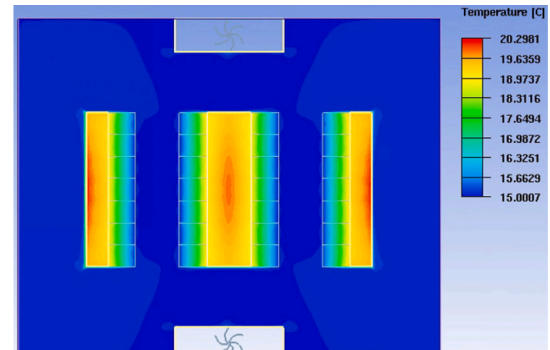
$$pPUE_{cooling} \approx \frac{1}{COP \cdot ECR} \quad (20)$$



(a) Temperature distribution at 1.2m of open aisle



(b) Temperature distribution at 1.2m of enclosed cold aisle



(c) Temperature distribution at 1.2m of enclosed hot aisle

Fig. 15. The temperature distributions of the horizontal plane for three models.**Table 4**

The detailed information of the open aisle model.

Equipment No.	Temperature / °C				Flow rate / $\text{m}^3 \text{s}^{-1}$	
	CRAC outlet	CRAC inlet	Rack inlet	Rack outlet	CRAC	Rack
1	15	19.3	18.2	19.8	11.4	13.5
2	15	19.3	17.7	19.4	11.4	13.2
3	–	–	17.7	19.4	–	13.2
4	–	–	18.2	19.8	–	13.5

where COP —the electric cooling efficiency.

Assuming that the air cooling temperature is constant, the energy consumption is proportional to the cooling capacity. The effect of

Table 5

The detailed information of the enclosed cold aisle model.

Equipment No.	Temperature				Flow rate	
	CRAC outlet	CRAC inlet	Rack inlet	Rack outlet	CRAC	Rack
1	15	19.3	15.2	19.3	11.4	5.2
2	15	19.3	15.2	19.3	11.4	5.1
3	–	–	15.2	19.3	–	5.1
4	–	–	15.2	19.3	–	5.2

Table 6

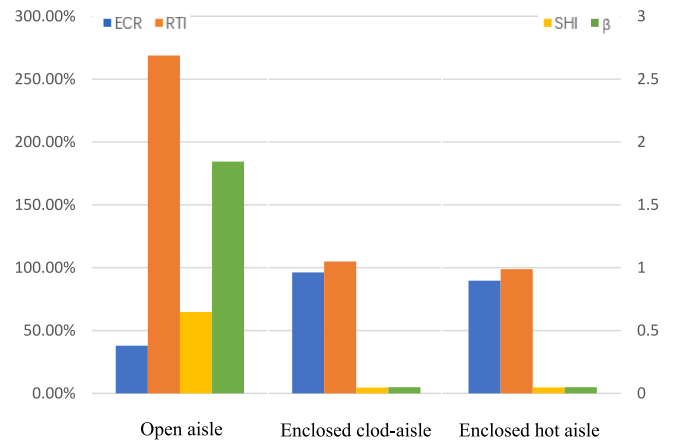
The detailed information of the enclosed hot aisle model.

Equipment No.	Temperature / °C				Flow rate / $\text{m}^3 \text{s}^{-1}$	
	CRAC outlet	CRAC inlet	Rack inlet	Rack outlet	CRAC	Rack
1	15	19	15.2	19.3	11.4	5.3
2	15	19	15.2	19.2	11.4	5.5
3	–	–	15.2	19.2	–	5.5
4	–	–	15.2	19.3	–	5.3

Table 7

Four indicators for the three models.

	Open aisle	Enclosed cold aisle	Enclosed hot aisle
ECR	37.98 %	96.22 %	89.64 %
SHI	0.6473	0.0465	0.0471
β	1.8438	0.0488	0.0494
RTI	268.75 %	104.88 %	98.78 %

**Fig. 16.** The value of the four indexes.

different ECR on PUEs is discussed as follows (Table 8).

For air-cooled data centers with a COP equal to 2, an ideal ECR is required to meet a PUE of less than 1.5. Generally, with a COP of 3, an ECR of 80 % is required to meet the relevant regulations. For the high efficiency of the second cooling system, or free cooling, whose COP is equal to or greater than 5, having a good ECR makes it possible to achieve a PUE approaching or even less than 1.2. This can achieve the same level of energy efficiency as a liquid-cooled data center. In addition, ECR can help to identify and solve problems in data centers that are not as good as the design.

6. Conclusions

This paper has investigated the cooling process of the computer room (i.e., the first cooling system) of an air-cooled data center and introduced

Table 8
The effect of different ECR on PUE under different COP.

COP=2		COP=3		COP=4		COP=5	
ECR	pPUE _{cooling}	ECR	pPUE _{cooling}	ECR	pPUE _{cooling}	ECR	pPUE _{cooling}
20%	2.5	20%	1.667	20%	1.25	20%	1
40%	1.25	40%	0.833	40%	0.625	40%	0.5
60%	0.833	60%	0.556	60%	0.417	60%	0.333
80%	0.625	80%	0.417	80%	0.3125	80%	0.25
100%	0.5	100%	0.333	100%	0.25	100%	0.2
PUE		---		---		---	
		--- PUE of data center: 1.4~1.5		--- PUE of data center: 1.3~1.4			
		--- PUE of data center: 1.2~1.3		--- PUE of data center: ~1.2			

the diagram method to propose the calculation method of the effective cooling rate. Then, the CFD simulation is used to conduct comparative studies of the four evaluation indexes for three airflow organization structures of data centers. The major conclusions are as follows:

- 1) To analyze the data center cooling system, the first cooling system, which acts directly on IT equipment, is divided into four phases, and the energy analysis diagram is developed.
- 2) Based on using the diagram to quantify the recirculation or bypass of each phase, a new index of effective cooling rate (ECR) is proposed for the first time to evaluate the efficiency of the first cooling system.
- 3) The four evaluation metrics of the three arrangement models are evaluated by the data from CFD simulation, and the results show that the enclosed aisle is a high-efficiency model with high values of ECR.
- 4) The relationships among ECR, PUE and COP are investigated, and for the data center investigated, every 20 % increase in ECR is equivalent to an increase in COP of 1, and a decrease in PUE of 0.1. ECR provides a way to couple a three-dimensional CFD simulation with a one-dimensional energy consumption model.

CRedit authorship contribution statement

Nan Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haowei Li:** Investigation. **Kaiwen Duan:** Investigation. **Wen-Quan Tao:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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